

# Bhavyasri Sanisetty

Backend Engineer — ML-Integrated API Systems & Secure Service Architecture

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- Published ML research — *Scientific Reports (Nature Portfolio, 2025)*
- Deployed ML inference backend service
- Engineered secure JWT-based MFA authentication system

## Flagship Production System

### Smart Agriculture ML Platform — Deployed Backend Service

- Designed and deployed an end-to-end **ML-powered decision platform** integrating **crop recommendation**, **weed**, **pest classification**, and **secure API access**.
- Benchmarked **20 ML/DL models** on a **2,200-record dataset (22 classes)** and deployed a hybrid ensemble (**Voting Classifier**) achieving **15–20% higher accuracy** than standalone baselines.
- Built image-classification pipelines covering **232 species** using **65k+ labeled images (ResNet50, CNN, ViT)**
- Architected modular **FastAPI**, **REST API inference services** (**model serialization**, **schema validation**, **structured JSON responses**) enabling reusable **production-grade ML APIs**.
- Secured and operationalized backend services via **JWT-based authentication** and **Railway deployment**, using **ngrok tunneling** and validating **live inference workflows**.

## Published Research

### Hybrid Deep-Learning Energy Load Forecasting — *Scientific Reports (Nature Portfolio), 2025*

- Led implementation of a hybrid **ANN + Transformer** forecasting framework integrating fuzzy preprocessing to model noisy multivariate energy load data and long-range temporal dependencies.
- Formulated data preprocessing and feature engineering pipelines to process real-world hourly energy consumption variables including electricity, gas, HVAC, and auxiliary signals.
- Benchmarked the hybrid model against traditional and standalone baselines (Random Forests, Decision Trees, Linear Regression) achieving **~15–20% reductions in RMSE and MAE** and strong generalization.
- Developed reusable training and evaluation workflows in **Python and PyTorch** enabling scalable experimentation across model variants.
- Validated model effectiveness on **real energy datasets** supporting improved load forecasting insights for grid planning.

## Additional Engineering Projects

### Multi-Factor Authentication Backend Service

*FastAPI · JWT · OTP · REST APIs*

- Engineered **JWT-based authentication** with **OTP verification** and **middleware-protected routes**.
- Implemented multi-step authentication flow (login → OTP verification → active session).
- Enforced **token expiry validation** and authorization checks across protected endpoints.
- Implemented role-protected routes with token validation and session state management.

### Prompt-to-UI Generator

*FastAPI · React · JSON API*

- Developed **FastAPI backend** converting natural-language prompts into structured UI component definitions.
- Designed **API contract** consumed by a **React frontend** for dynamic component rendering.
- Implemented asynchronous processing and **request validation** to safely handle malformed inputs.

## Professional Experience

### Salesforce Platform Configuration

May 2024 – Jun 2024

- Configured CRM objects, validation rules, and **workflow automation**.
- Implemented business process logic simulating **enterprise data operations**.

## Technical Skills

**Backend & APIs:** FastAPI, Flask, REST API Design, JWT Authentication, Request Validation

**ML & AI:** PyTorch, scikit-learn, Time-Series Modeling, Deep Learning (ANN, CNN, RNN/LSTM, Transformers), Pandas, NumPy

**Databases:** MongoDB, SQL

**Infrastructure & Tools:** Git, Linux, Railway, ngrok

**Languages:** Python, JavaScript

## Education

B.Tech Computer Science & Engineering — SRM University (2025)